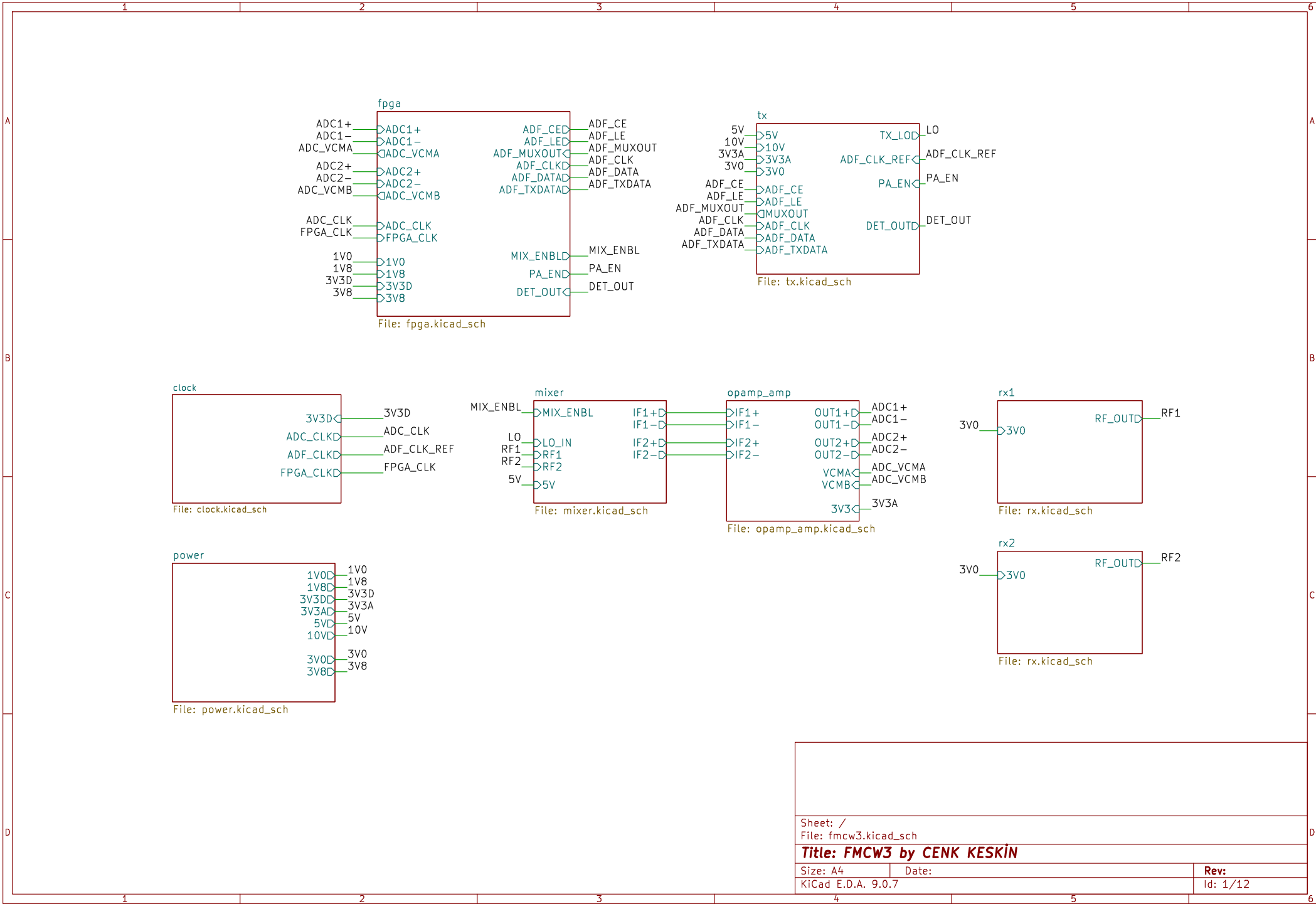
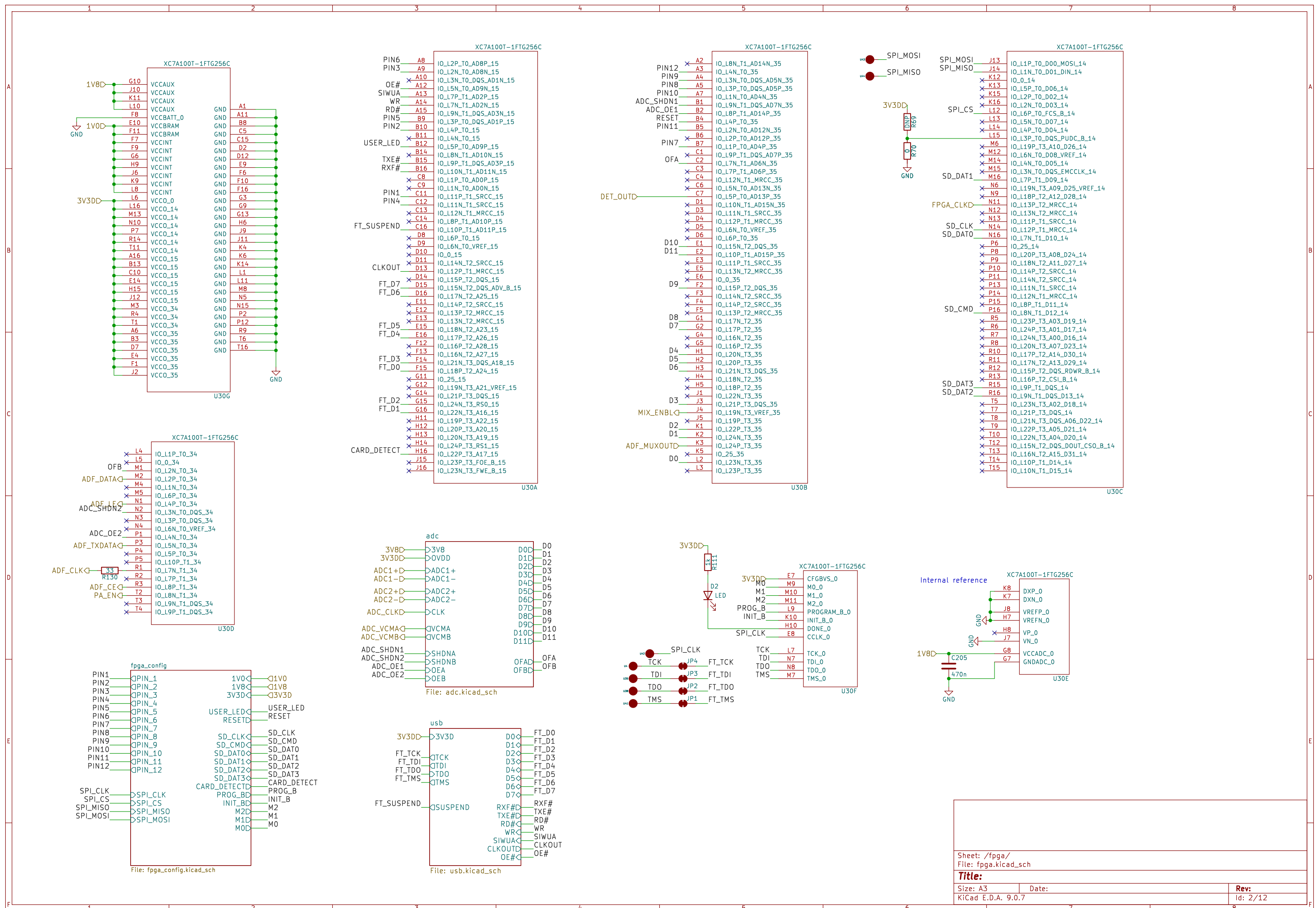
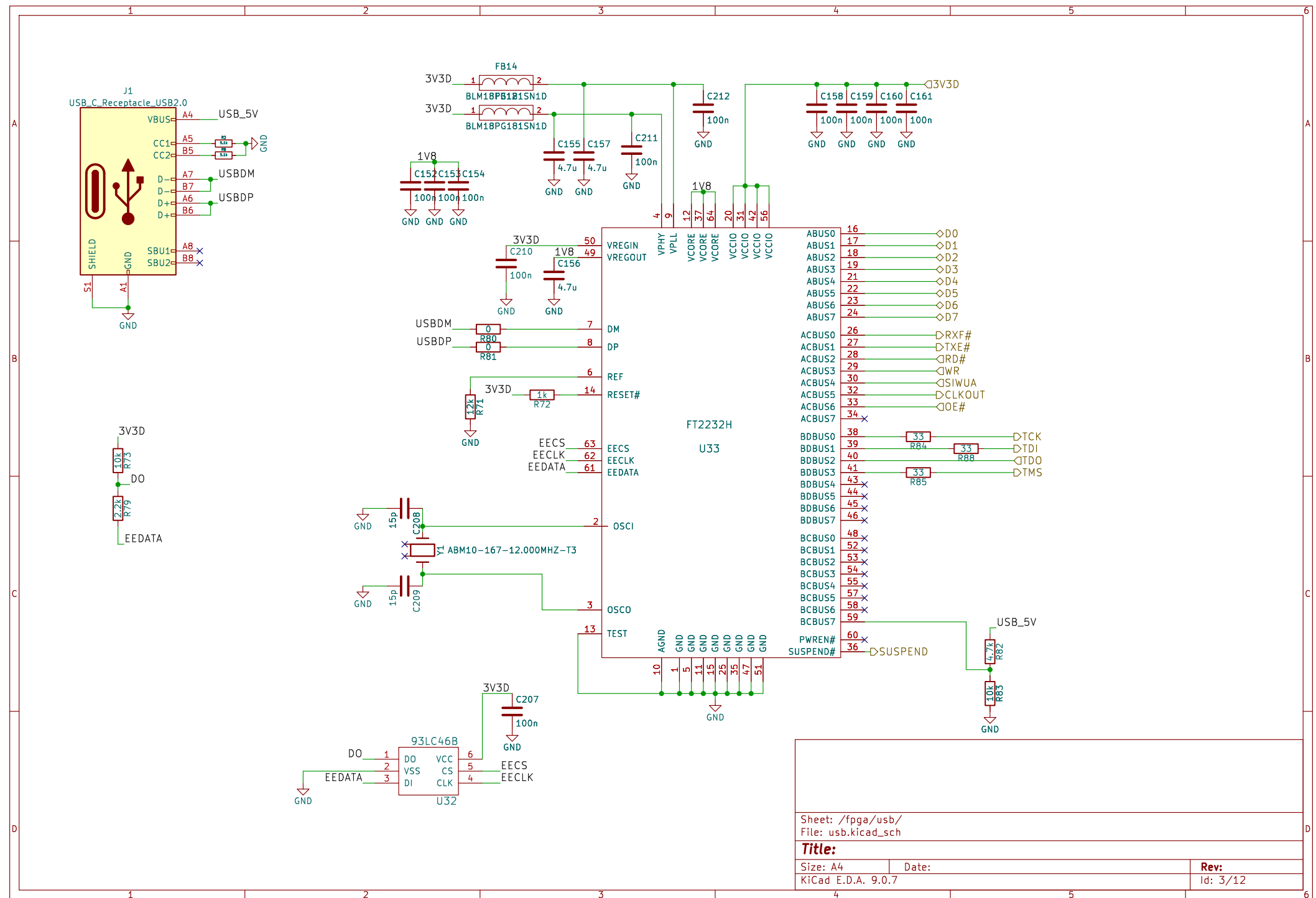
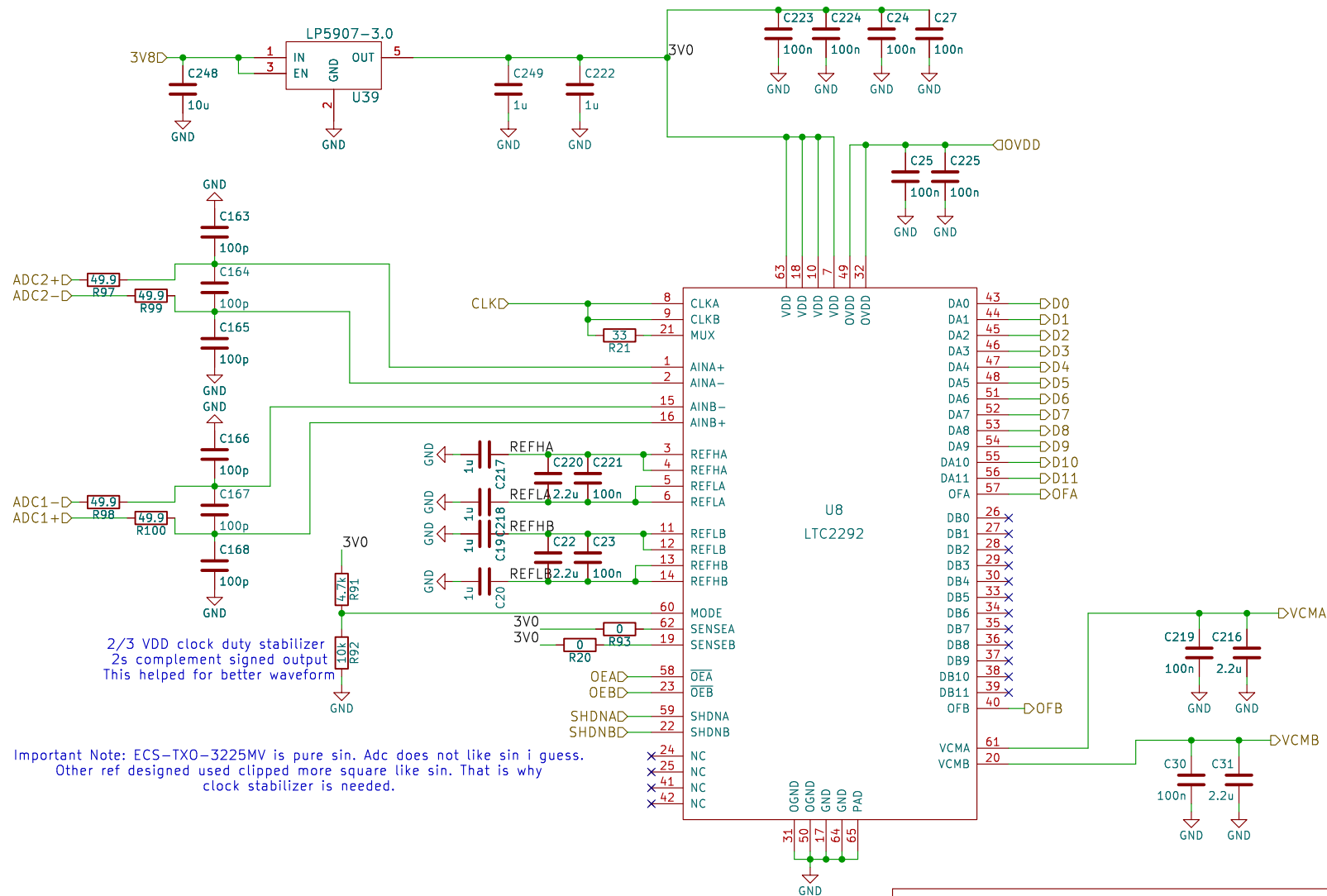










Sheet: /fpga/adc/  
File: adc.kicad\_sch

# Title:

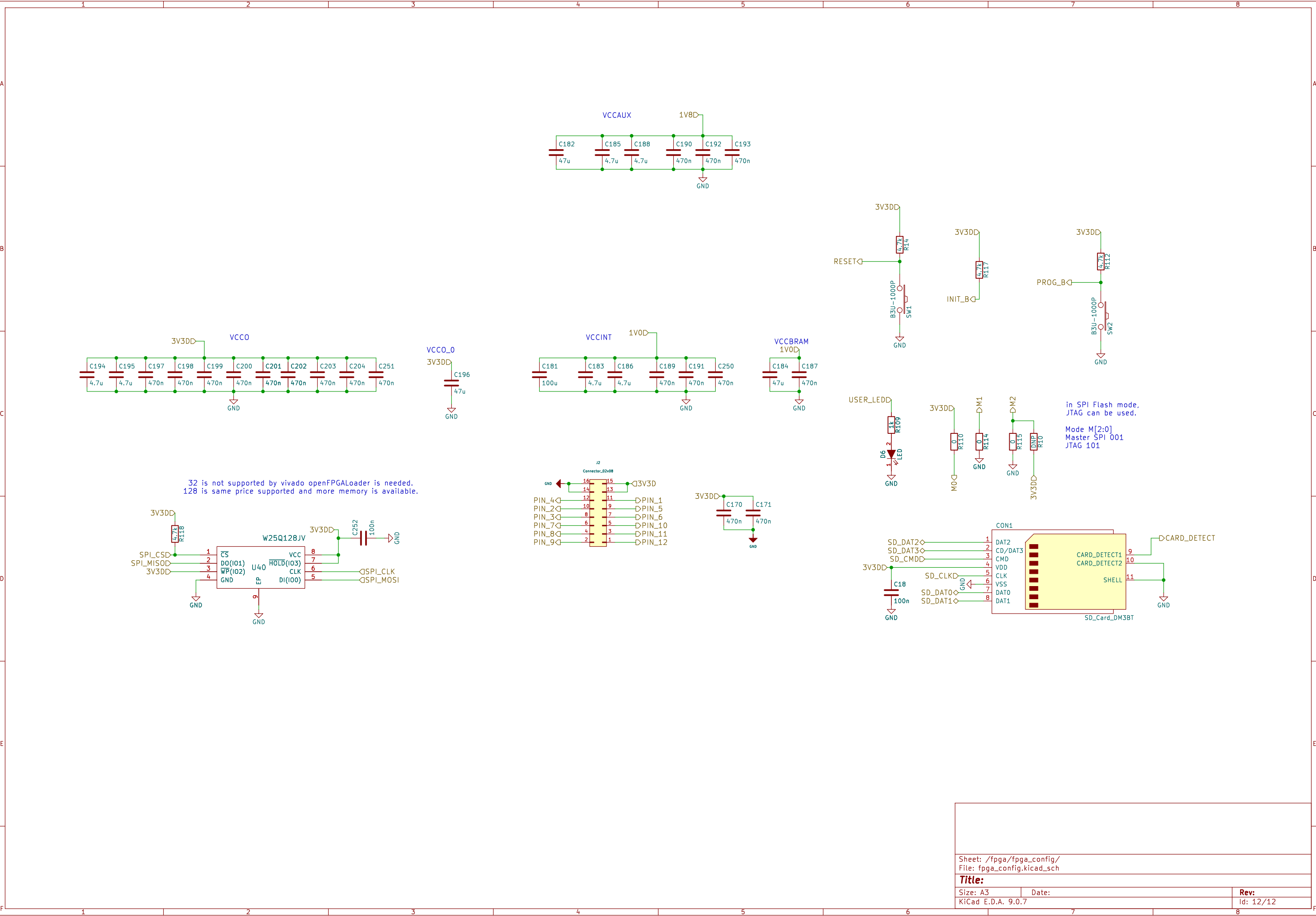
Size: A4

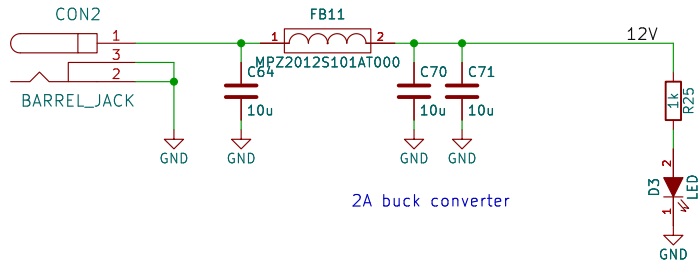
Date:

KiCad E.D.A. 9.0.7

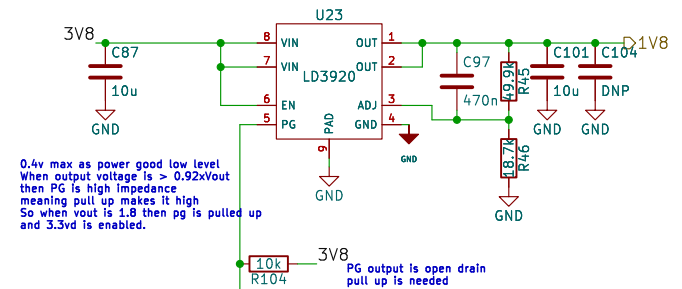
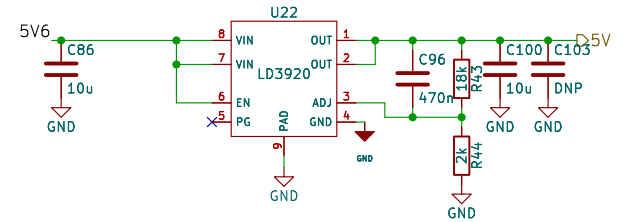
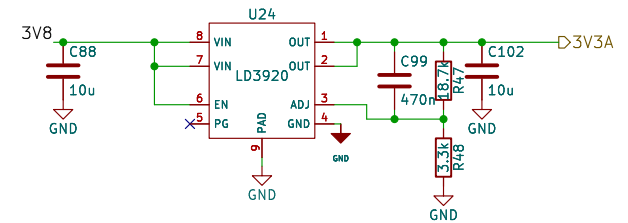
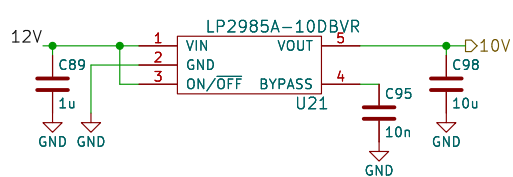
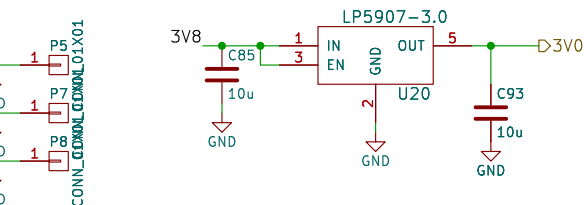
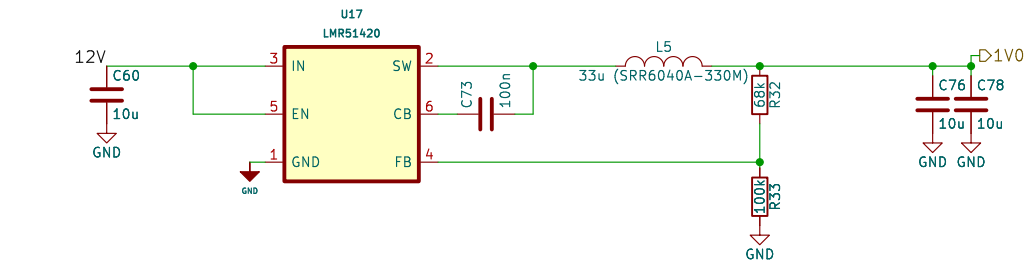
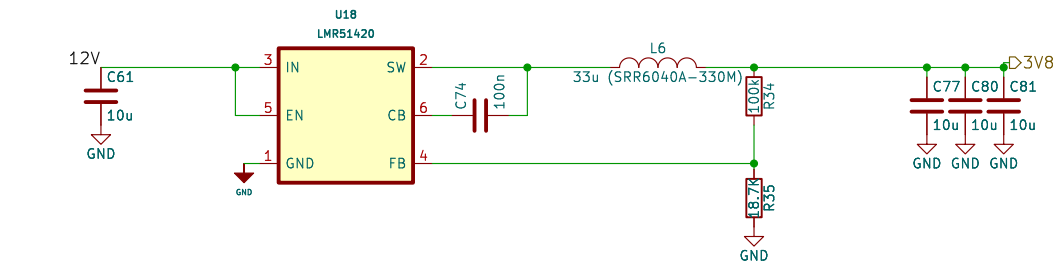
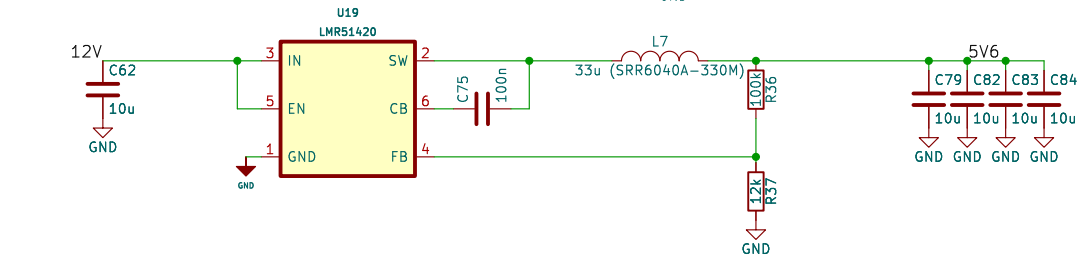
Rev:

Id: 4/12

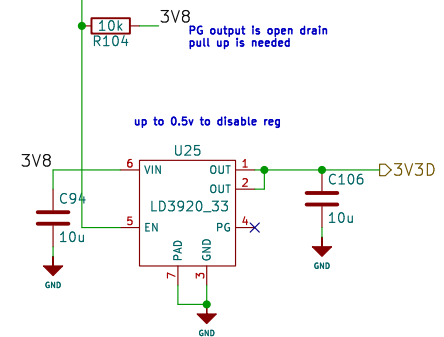




In the Artix-7 data sheet it states that the recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO. 1v0 first 1v8 then 3v3 is the sequence so use power good



0.4v max as power good low level  
When output voltage is > 0.92xVout  
then PG is high impedance  
meaning pull up makes it high  
So when vout is 1.8 then pg is pulled up  
and 3.3vd is enabled.



Sheet: /power/  
File: power.kicad\_sch

Title:

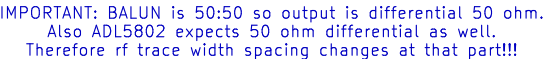
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Date:

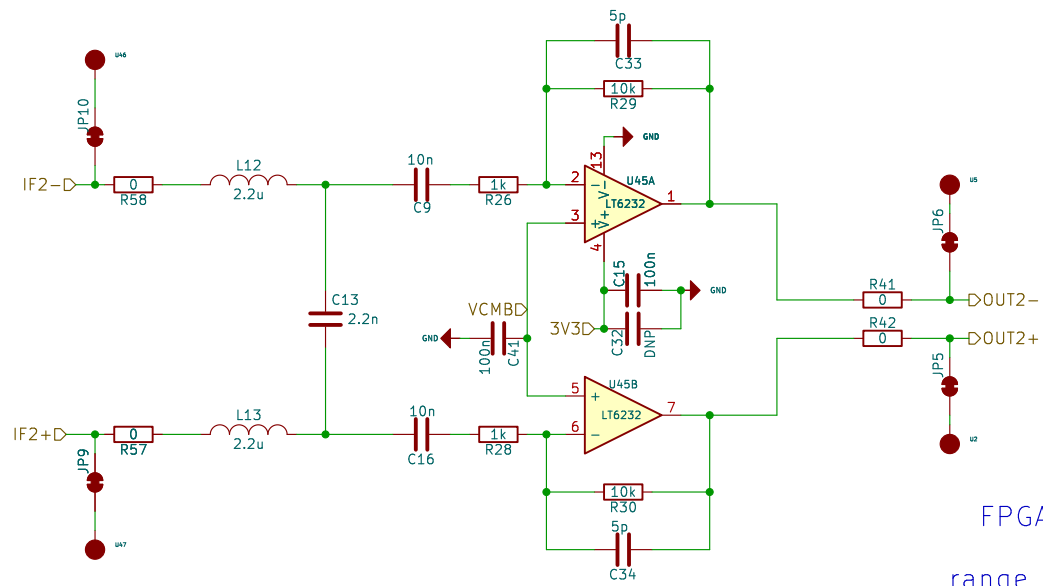
KiCad E.D.A. 9.0.7

Rev:

Id: 5/12

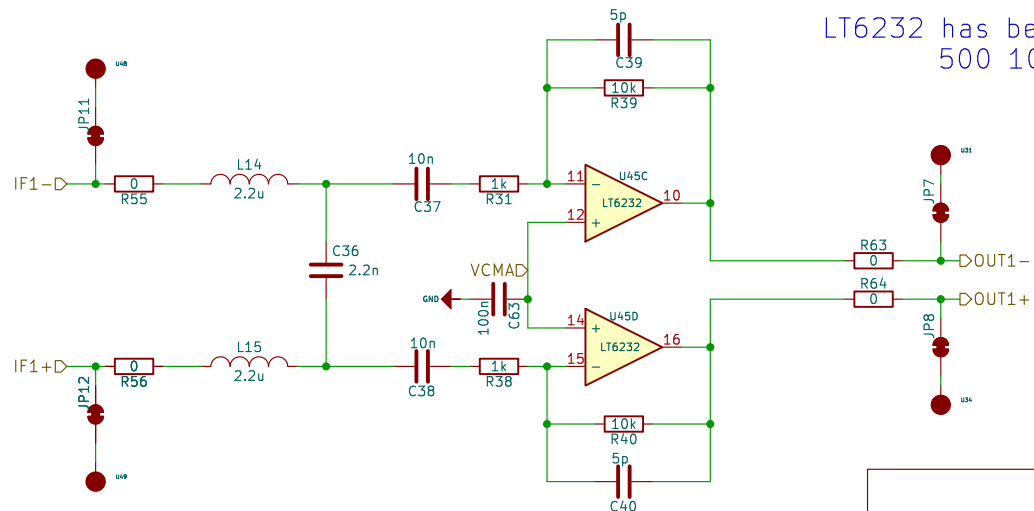


Id: 6/12



FPGA FIR Filter works perfectly.  
Second stage for HPF  
range compensation filter is removed

LT6232 has better noise performance as  $1.1\text{nV}/\sqrt{\text{Hz}}$   
500 10K has less noise contribution



Sheet: /opamp\_amp/  
File: opamp\_amp.kicad\_sch

**Title:**

Size: A4

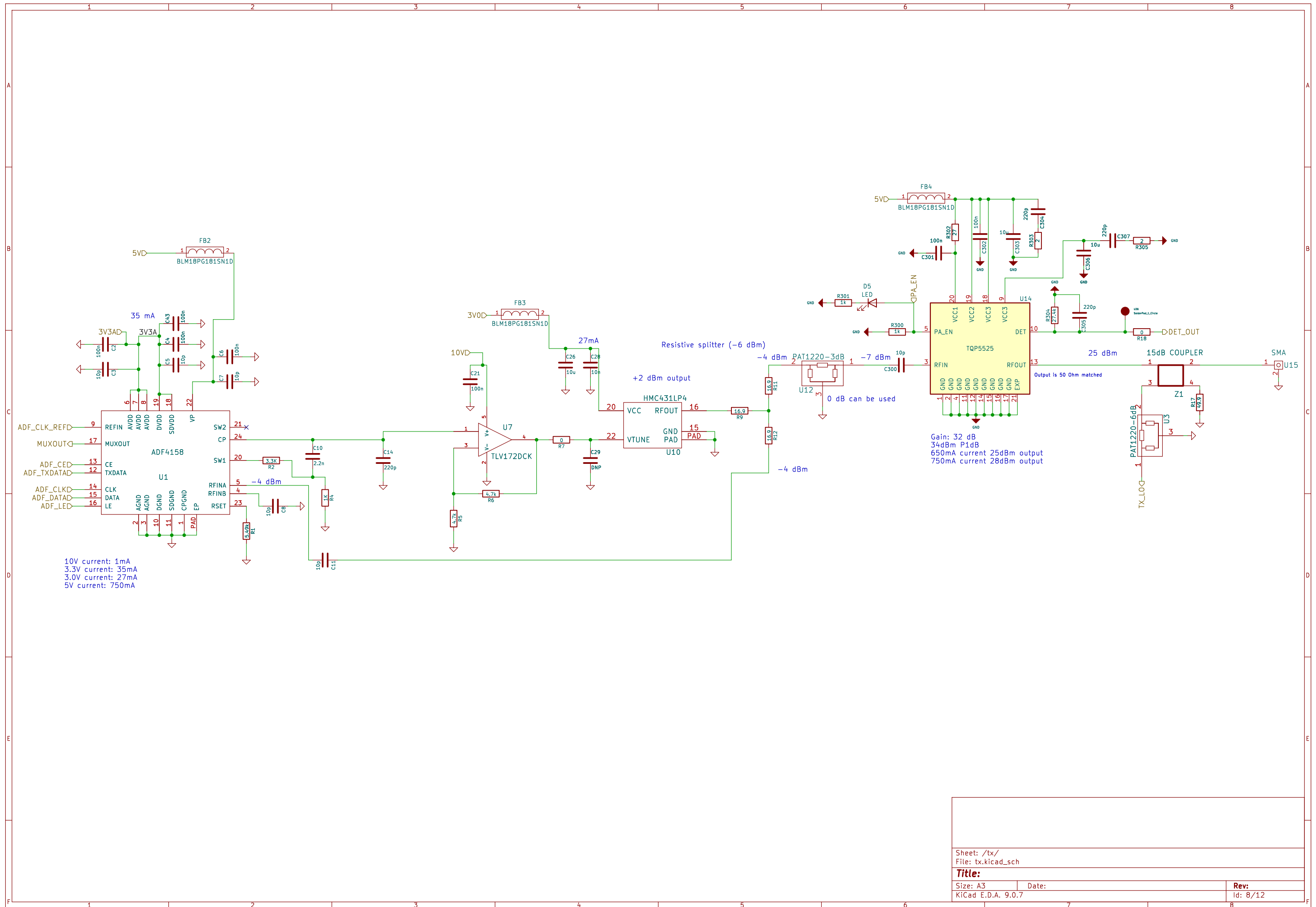
Date:

**Rev:**

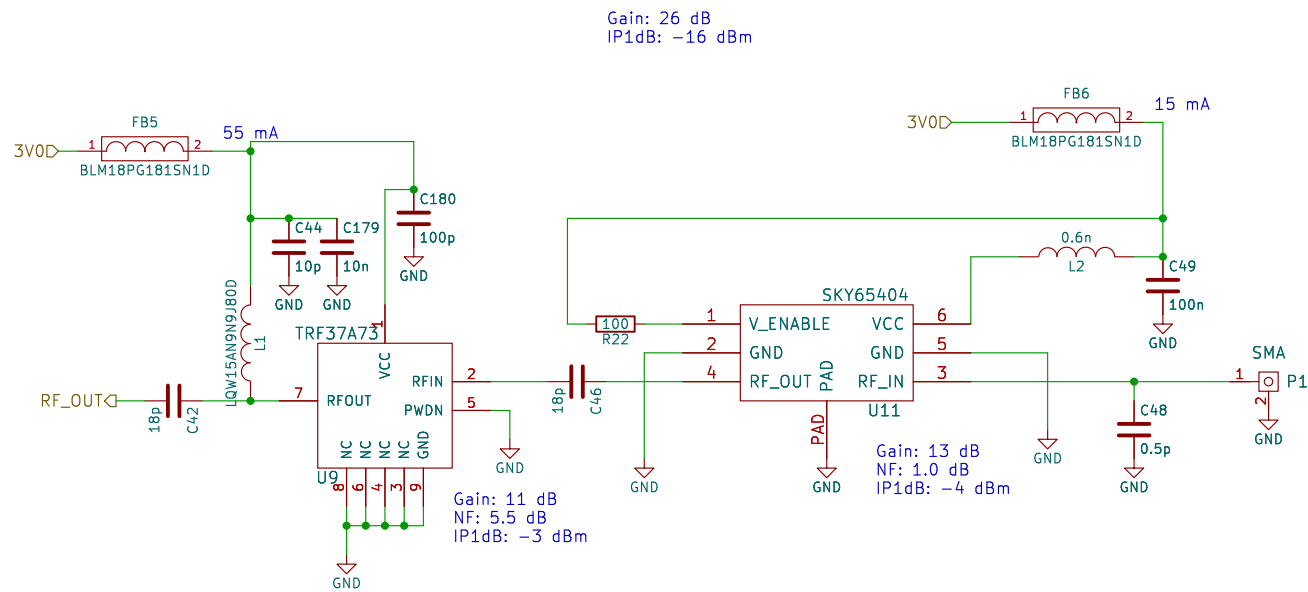
KiCad E.D.A. 9.0.7

Id: 7/12





10V current: 1mA  
3.3V current: 35mA  
3.0V current: 27mA  
5V current: 750mA



3.0V current: 70mA

Sheet: /rx1/  
File: rx.kicad\_sch

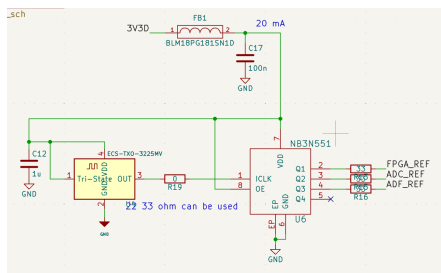
**Title:**

Size: A4  
KiCad E.D.A. 9.0.7

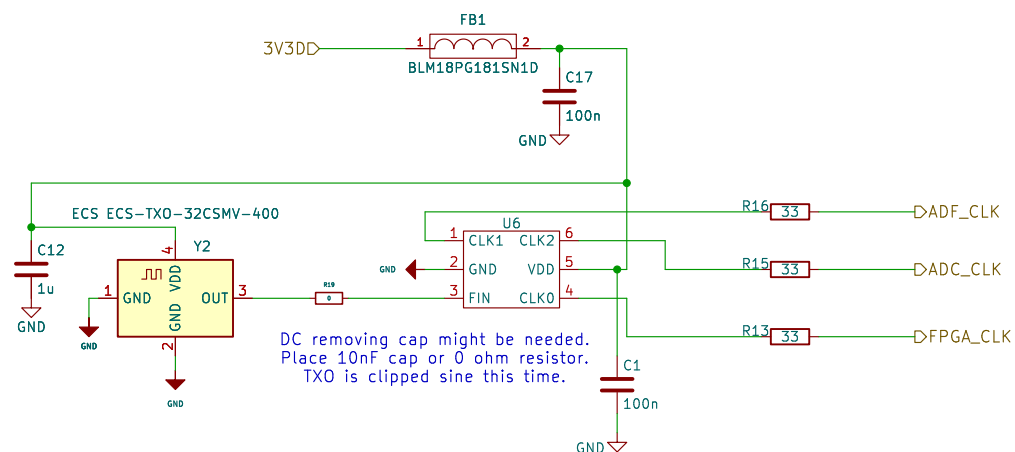
Date:

Rev:  
Id: 9/12





This is the prev design:  
NB3N551 is obs.  
Also this txo is generating pure sin.  
ADC might need cmos like input not sin.  
That could be the reason of duty stabilizer.



Sheet: /clock/  
File: clock.kicad\_sch

# Title:

Size: A4  
KiCad E.D.A. 9.0.7

Date:

Rev:  
Id: 11/12